

PATH PROTOCOL – WORKING ANALYSIS CHECKLIST

Protocol of Analysis of senTiment in Health | Source: He et al. (2021), JAMIA 28(6):1125–1134

Instructions: For each PATH item, document how your study addresses or departs from the recommended design choice. Use the 'Our Dataset Details' column to record decisions, rationale, and validation outcomes.

DIMENSION	ITEM CODE	DESIGN CHOICE	DESCRIPTION / GUIDANCE	OUR DATASET DETAILS
<i>Platform Selection</i>	PS1	Platform description	Describe the social media platform studied and its user/content characteristics relevant to the research question.	The dataset used in this study was developed using AI-assisted synthetic data generation techniques rather than directly collecting user data from social media platforms. To ensure realistic language patterns, publicly available communication styles commonly observed on Reddit and X (formerly Twitter) were considered during dataset design. This approach enabled the creation of clinically relevant mental health text samples while avoiding the use of real user content, protecting privacy, and addressing ethical concerns related to personal data collection.
	PS1-A	Justification for platform selection	Explain why the chosen platform is the most informative for the study, compared to other available options.	Although the dataset was generated synthetically using AI-assisted methods, communication patterns commonly observed on social media platforms such as Reddit and X (formerly Twitter) were considered during its development. These platforms are particularly informative for mental health research because they contain large volumes of first-person narratives, emotional expressions, and discussions related to psychological well-being. Compared to many other social media platforms, they provide richer text-based content that captures users' thoughts, experiences, and symptom descriptions in a natural language format, making them valuable references for developing and evaluating sentiment analysis and mental health classification models while maintaining privacy through synthetic data generation.

Reference

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<i>Data Curation</i>	DC1	Data retrieval method	Describe the strategy used to retrieve social media data (keywords, hashtags, user surveys, geo-tags, etc.).	No social media data was directly retrieved, scraped, or collected for this study. Instead, the dataset was developed entirely through AI-assisted synthetic data generation. To ensure realistic language patterns and symptom representations, an observational review of publicly accessible mental health discussions and support communities on platforms such as Reddit and X (formerly Twitter) was conducted. This review focused exclusively on identifying common linguistic structures, emotional expressions, symptom descriptions, and communication styles associated with Depression, Anxiety, PTSD, Social Anxiety Disorder, and Suicidal Ideation. No specific posts, comments, user profiles, hashtags, geo-tags, personal identifiers, or private information were downloaded, copied, stored, or included in the dataset. Consequently, traditional social media data retrieval strategies such as keyword extraction, hashtag collection, user surveys, or geolocation filtering were not employed, thereby ensuring compliance with privacy and ethical research considerations.
	DC2	Relevance & comprehensiveness determination	Report the method used to assess whether retrieved data are relevant and comprehensive (e.g., iterative keyword refinement, manual review of sample).	Since no social media data were directly retrieved, traditional relevance assessment methods such as keyword refinement, hashtag filtering, or data retrieval validation were not applicable. Instead, dataset quality and comprehensiveness were ensured through an iterative synthetic data generation process. AI-generated samples were manually reviewed and refined to verify that they accurately reflected clinically recognized symptoms, emotional expressions, and linguistic patterns associated with Depression, Anxiety, PTSD, Social Anxiety Disorder, and Suicidal Ideation. Multiple iterations of generation and review were conducted to ensure diversity in language, symptom severity, and expression styles while minimizing redundancy. Additionally, publicly accessible mental health

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				discussions on online communities were observed solely to validate the realism and representativeness of the generated narratives without collecting, storing, or reproducing any user-generated content.
	DC3	Differentiation by content creator	Decide whether to treat data differently based on creator type: laypersons, HCPs, organizations, bots, pharma, etc.	Creator type was not considered as a factor in dataset construction because no social media content was directly collected or analyzed at the individual post level. The dataset was generated synthetically using AI-assisted methods based on general linguistic and symptom-expression patterns observed in publicly accessible mental health discussions. Consequently, no distinction was made between content originating from laypersons, healthcare professionals, organizations, automated accounts, or commercial entities. The focus of the study was on generating realistic mental health-related textual expressions rather than examining differences between specific content creators or user groups.
	DC4	Handling of special data types	Report how hashtags, emojis, images/videos, and hyperlinks were handled at preprocessing (retain, remove, or substitute).	As the dataset consisted entirely of AI-generated textual narratives, multimedia elements such as images, videos, and external hyperlinks were not included. Hashtags and emojis were also excluded during the generation and preprocessing stages to maintain a consistent clinical narrative style and reduce noise in the text. Consequently, no substitution or normalization procedures were required for these elements. The final dataset contained only plain-text symptom descriptions and self-report statements suitable for sentiment analysis and mental health classification tasks.
	DC4-A	Justification for special data handling	Explain the rationale for the chosen approach to special data types and its implications for results.	The decision to exclude special data types such as hashtags, emojis, images, videos, and hyperlinks was made to maintain a standardized and clinically oriented text corpus focused on linguistic indicators of mental health conditions. Since the objective of the study was to develop and evaluate a text-based

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				sentiment and mental health classification model, retaining only plain-text narratives helped reduce noise and variability that could arise from platform-specific features. This approach enabled the model to focus on symptom descriptions, emotional expressions, and contextual language patterns. However, the exclusion of non-textual and social media-specific elements may limit the model's ability to capture certain emotional cues commonly conveyed through emojis, hashtags, or multimedia content in real-world social media environments.
<i>Sentiment Analysis Method</i>	SAM1	Sentiment analyzer description	Identify and describe the sentiment analyzer or machine learning model used.	A two-stage classification pipeline was used: Stage 1 – Azure Sentiment Analysis (Microsoft Azure Cognitive Services): A cloud-based transformer-based NLP service trained on large multilingual corpora. For each input, it returns a sentiment label (Positive / Neutral / Negative) accompanied by probability confidence scores ($P_{\text{positive}} + P_{\text{neutral}} + P_{\text{negative}} = 1$). Only inputs labelled Negative proceed to Stage 2. Stage 2 – Azure Custom Text Classification (Azure Language Studio): A managed multi-class classifier trained on the study's synthetic dataset. It categorises negative-sentiment text into one of five predefined classes: Anxiety, Depression, PTSD, Social Anxiety Disorder, Suicidal Ideation and Behaviour. The predicted class is: $\hat{c} = \text{argmax } P(c_i x)$ over all categories $c_i \in C$.
	SAM1-A	Justification for tool selection	Explain why the chosen analyzer is appropriate for the study data and context.	Azure Sentiment Analysis was selected because: (a) it uses contextual transformer models rather than lexicon-based matching, making it more sensitive to nuanced emotional language; (b) it is scalable and integrates natively into the Azure cloud ecosystem used by the Android application;

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				(c) it provides probabilistic outputs that enable threshold-based routing to Stage 2. Azure Custom Text Classification was selected because: (a) it allows fine-grained domain-specific categorization beyond the three-class sentiment output; (b) it supports custom label sets directly aligned to the five target mental health categories; (c) it automates feature extraction, training, and evaluation within a managed environment, ensuring reproducibility; (d) its low inference latency (20 ms vs 420 ms for BERT-base) supports real-time mobile deployment. The hierarchical design reduces computational overhead by routing only negative-sentiment inputs to the more resource-intensive classification stage.
	SAM2	ML feature description (if applicable)	For ML-based models, describe the features selected to train the model (bag-of-words, embeddings, linguistic features, etc.).	Azure Custom Text Classification is a managed service; internal architectural details (embedding structures, hidden layers, optimization strategy, model weights) are not exposed to end users. The classification process is therefore described in terms of input-output behaviour. The custom model is informed by a BERT-based architecture (12-layer transformer encoder, 768-dimensional hidden states) used during independent benchmarking. Key features implicitly learned by the model include: - Contextual word and sub word embeddings capturing long-range dependencies within the input text; - Sarcasm-sensitive representations via a contrastive learning objective distinguishing literal from sarcastic expressions;

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				Dropout (rate = 0.2) for regularization; Focal loss ($\gamma = 2$) to address class imbalance; learning rate 5e-5 with linear warmup over 1,000 steps.
	SAM2-A	ML feature justification (if applicable)	Justify the feature selection process and explain how features relate to the study's analytical goals.	Contextual embeddings (BERT-style) were chosen over bag-of-words or TF-IDF features because mental health language is heavily context-dependent. The phrase "I don't want to be here" has different implications depending on surrounding text; a contextual encoder captures these dependencies. Focal loss was selected to prevent the model from being dominated by high-frequency categories; class-weighted training ensures that less common but clinically critical categories (e.g., Suicidal Ideation) are not systematically under-classified.
	SAM3	Pre-study validation	Validate the selected sentiment analyzer against the study data before full application, even if used in prior studies.	Both components were validated prior to full application: Azure Sentiment Analysis was validated against a 1000-sample manually labelled dataset (200 per category) including standard statements, sarcastic expressions, and emoji-laden text. Overall accuracy: 89% on explicit expressions; 78% on sarcastic comments; 65% precision on positive-emoji-masked negative sentiment. These results confirmed the need for Stage 2 and guided routing logic. Azure Custom Text Classification was validated using an 80/20 train-test split on the primary synthetic dataset, achieving overall Precision, Recall, and F1-score of 96.97%. A secondary external validation was then performed on the 125 -item clinically annotated dataset (see SAM3-A), generating an independent confusion matrix and expert-agreement analysis.

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	SAM3-A	Annotated data for validation/training	Obtain manually annotated data for validation or training; prefer study-specific data over external corpora.	Three annotation layers were produced, all study-specific: (1) Primary training + testing dataset: Entirely synthetic, manually constructed samples representing all five categories. Labels assigned by the construction team based on observational linguistic analysis. Used in 80/20 train-test split for Azure Language Studio training. (2) Co-author reviewed dataset: A representative subset of synthetic samples was independently reviewed by two co-authors with mental health domain knowledge to verify label consistency and linguistic realism before training. (3) Clinically annotated external validation dataset: 125 synthetic samples (25 per category) independently annotated by qualified mental health professionals (psychologists). This dataset was not used in training and served exclusively as an external validity benchmark, generating a confusion matrix to compare model predictions against expert judgement. No publicly available clinical datasets were used for training, as existing resources (therapy transcripts, PHQ assessments) did not align with the five-category label structure and text-classification task.
	SAM3-B	Multiple coders for annotation	Involve multiple independent coders for annotation to ensure quality, given the subjectivity of health sentiment.	Multiple independent coders were involved across two stages: Stage 1 (Co-author review): Two co-authors with mental health domain knowledge independently reviewed representative samples from each of the five categories. Each reviewer assessed linguistic realism, category appropriateness, and label consistency independently before comparing results. Discrepancies were not found.

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				Stage 2 (Clinical annotation): The 125-item external validation dataset was annotated by qualified mental health professional (psychologist). Independent annotation was performed prior to any cross-reviewer discussion, ensuring that initial labels reflected individual clinical judgement.
	SAM3-B-1	Inter-rater reliability reporting	Quantitatively assess and report inter-rater reliability (e.g., Cohen's κ , percentage agreement) when multiple coders are used.	Formal quantitative IRR statistics (e.g., Cohen's κ) are not explicitly reported in the manuscript as a named metric. This is acknowledged as a limitation of the current reporting. What is reported: agreement-based consensus resolution for the co-author review stage (discrepancies were not found), and model-vs-expert agreement operationalized through the independent confusion matrix generated from the clinically annotated validation dataset.
	SAM3-C	Quantitative validation metrics	Report validation results using established metrics: accuracy, F-score, precision, recall, AUC-ROC.	Full quantitative metrics were computed and reported for both validation stages: Primary evaluation (80/20 split, Azure Language Studio): Overall Precision = 96.97%, Recall = 96.97%, F1-score = 96.97%. Class-level F1-scores ranged from 0.94 to 1.00 across all five categories. Confusion matrices presented for primary held-out test set. Secondary evaluation (clinically annotated 125-item dataset): Independent confusion matrix generated comparing model predictions against expert annotations. Class-level precision, recall, and F1-scores analyzed with particular attention to balanced performance and minimization of false negatives for high-risk categories (Suicidal Ideation: F1 = 0.97). Statistical validity: 95% confidence intervals computed using the Clopper-Pearson method. At $n = 1,200$ and observed accuracy of 95%, the 95% CI was 93.2%–96.5%.

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				Note: AUC-ROC was not separately reported; this could be added for binary one-vs-rest analyses per category in future reporting to fully satisfy this criterion.
	SAM4	ML model comparison (if applicable)	Evaluate multiple competing ML models (e.g., SVM, Naive Bayes, logistic regression) before selecting the final model.	A formal benchmarking comparison was conducted against four models using identical training data and hardware (NVIDIA A100 GPUs): • Logistic Regression: Precision lower by ~17.3% F1 gap vs custom model; training time < 1 min; inference latency not reported. • BERT (base): Comparable accuracy but inference latency of 420 ms vs 20 ms for custom model; training time not directly comparable given managed Azure environment. • RoBERTa (large): Comparable accuracy to custom model but training time 6 hrs 45 min vs 5 min 10 sec; inference latency 680 ms — prohibitively slow for real-time mobile deployment. • Custom Azure model (selected): The model achieved overall F1-score = 96.97%, Precision = 96.97%, and Recall = 96.97% on the held-out test set. Training time ~5 min. Superior balance of accuracy, latency, and training efficiency. The custom Azure model was selected on the basis of best overall performance, lowest inference latency (critical for real-time mobile use), and most efficient training pipeline. SVM and Naive Bayes were reviewed in the literature survey but not individually trained on the study dataset; their omission from the primary benchmark is noted as a minor limitation.

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Legend:

- **HCP** = Healthcare Provider
- **IRR** = Inter-Rater Reliability
- **ML** = Machine Learning
- **AUC-ROC** = Area Under the Receiver Operating Characteristic Curve

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Certification

I/We certify that the research design choices documented in this PATH protocol work file accurately reflect the methodological decisions made in the study, and that all entries in the 'Our Dataset Details' column have been reviewed and verified by the undersigned.

PRINCIPAL INVESTIGATOR / RESEARCHER	SUPERVISOR	CLINICAL PSYCHOLOGIST
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